

EE/CprE/SE 491 WEEKLY REPORT 3

10/5 – 10/9 Group: 28

Project title: Biomimetic Prosthesis with Precise Recreation of Hand Movement

Client &/Advisor: Erik Hrysyk, Meng Lu

Team Members/Role:

Reid Post – Prosthetic Researcher

Redwan Elshafiey – Sensors and Control Researcher

Logan Ward – EMG and Signal Processing Researcher

Adriana Cedillo Sanchez – Anatomy Researcher

Jacob Page – EMG and Materials Researcher

Erik Hrysyk – EMG/powertrain Researcher

Quinn Snyder – PCB/Electronics Lead

Weekly Summary

This week was another week of research and prep work. We as a team decided to order some cheap EMG sensors to get more hands-on training with signal processing. The novelty issues from last week were solved and our projects scope is becoming more defined. It is likely to assume that we need to split into 3 teams. A pressure team that helps to determine what we are going to do with the pressure and temp sensors, including how to implement the sensors and how to alert the user, an EMG team that is focused on utilizing the EMG's in order to provide commands to the motors. For this week, the EMG team should look into an open sourced EMG in order to avoid the hit on the teams budget, and a Hand team that is working on how to procure a hand, or buy and industry hand and also how to integrate that in with the controls from the EMG team. We have also toyed with the ideas of creating our own PCB for the EMG sensors to work around the budget hit on the EMG's, meaning that we are hoping to find an open sourced model for the EMG that could provide a workaround for another group's project.

Past week accomplishments

- We confirmed more of the scope about what the project, ordered cheap EMG sensors, reached out to staff looking for sensors,
- Reached out to Dr. Sarkar, and determined that it would not be wise to seek mentorship in ML, as that is not the direction we would like the project to go into.

- We have identified a few other contacts to reach out to for an EMG arm band, and have reached out to them
- Determined that the main focus is to acquire EMG sensors to maintain a buffer on the groups total budget
- Looked into IRB documentation so that this project can legally utilize EMG data for robotic operation
- Found some preexisting EMG training data that we can utilize for our project.

○ **Pending issues**

- Confirm project scope
- Buy EMG sensors (set up testing environment)
- Determine microcontroller
- Figure out underactuation strategies
- We have a large amount of I/O ports
- Determine whether past group's robotic hand can be adopted by our group
- We do not have an EMG band, and it may wipe out our budget
 - If we are buying an EMG band, we will have to reduce budget for everything else
 - If we can borrow an EMG, are we then having to worry about any other legal issues, or will we only have limited access to this band
 - We are looking into building our own custom EMG PCBs
- We do not have a project timeline, largely because we do not have the actual project scope defined

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u> <i>(Quick list of contributions. This should be short.)</i>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Logan Ward	Working on project website, figuring out testing environment for EMG sensors.	6	18
Reid Post	Looked into underactuation strategies, communicated with group as feedback came back in from other potential mentors	6	18
Redwan Elshafiey	Working on our project website, researching applicable training data, researching pressure sensors	6	18

Erik Hrysyk	Further EMG research, Signal processing study, comparison to other EMG controlled hands, reached out to others to try finding materials	6	18
Jacob Page	Movement and control research, sensor research, IRB Waiver submission, timeline creation	6	18
Adriana Cedillo	Brainstorming, researching anatomy and hand movement, researching for open source	6	18
Quinn Snyder	Research PCB methods, specifically EMG for now	6	18

- **Comments and extended discussion** (*Optional*)
- **Open Action Items**
 - Find open-sourced Github EMG model for gesture recognition
 - Buy/Find EMG (multi-channel)
 - Figure out how motor will interface with the ML and the EMG
 - See how the ML can be exported into the microprocessor
 - Brainstorm potential add-ons to existing hand
 - Optical mouse
 - Set up GitHub page for all data acquisition, signal analysis, machine learning, etc.
 - Provide a preliminary timeline for the project
 - Apply to the IRB for waiver and permissions
 - Look for an industry product hand that we can buy/open source
 - Provide a semester timeline
 - Determine IRB through Iowa State Website
 - Start assigning tasks to specific members and splitting into subgroups
 - Discuss PCB plans with possible mentors for PCB design help
 - Ensure that a custom PCB is realistic, reasonable, and worth the energy
 - Ask Rover who the customer was for last year's Hand project to determine who owns the arm
 - Figure out how to buy the EMG sensors to begin preliminary testing (likely need 10 kHz)
 - Determine if we are building our own EMG sensors, using last year's work, or are we buying our own
 - Determine what types of pressure/temp sensors that we would like to use
 - Determine where to place the electrode for the highest conductivity in the arm
 - Look into the different strategies for temperature sensing, because fast response is important for a large thermal mass like the hand

- Compare sensors in the market and the advantages and disadvantages of each one.
- Look into ordering a microcontroller as a backup plan.
- **Plans for the upcoming week**
 - Logan – Put onto EMG team to look into designing our own EMG to save money and so we have control over our design. Meet Quinn's TA with Quinn to inquire about future help with PCB design.
 - Reid – Look into documentation for earlier groups to determine if we need to buy a commercial EMG/hand and what strategies are possible if we can procure any part of an old project. Create and populate a research document highlighting possible market options for hands that have SDK capabilities.
 - Erik – Research EMG signal processing, reach out to kinesiology department, order EMG sensors/practice with sensor processing.
 - Adriana - Research for open source hand with in given budget. Finding best fit for project applications.
 - Quinn – Understand possible custom EMG strategies. Research what it will take to build our own, and how we might go about doing so. Find online resources that may have open-source files for us, or something we could redraw ourselves in KiCAD.
 - Redwan – Analyzing EMG training data, setting up the Git for our project, sorting out our project website, looking into the pressure and temperature sensors that we can ideally use for our hand
 - Jacob – Research different kinds of pressure and temperature sensors, look into the specs for a human grip, pick out the type of sensors we want to use. Send an email to the institutional review board to submit waiver to use human data. Create a timeline for the rest of the semester.
- **Summary of weekly advisor meeting**
 - Acquired IRB waiver to sign for EMG testing
 - Who was customer for last year's project? (used to determine who has the arm that we can use)
 - Go through ETG to purchase items for our project
 - Machine Learning Github repository has been found
 - Find sensors to use for pressure and temperature
 - Be able to estimate normal pressure range for a human hand and adapt that to our sensors

- How much pressure is required to pick something up? (hard object, fragile object, soft object)
- Professor Lu found an EMG from SparkFun
- Talked about potential strategies for temperature and pressure sensing
- Talked about having three teams; sensors, EMG, and the hand itself
- Talked about creating our own PCB for the EMG sensor, look for open source models